

Exercise-Introduction to Databases

Q1)Problem Statement: There can be multiple customers, who can place multiple orders on the site. Now a sales person can handle these orders will distribute into multiple sales persons (One order will be assign to one salesperson only). So a sales person can have multiple orders of multiple customers Create Database

```
mysql> create database db;
Query OK, 1 row affected (0.01 sec)

mysql> use db
Database changed
mysql> 
```

Problem Understanding:

1. There are multiple customers
2. Each customer can place multiple orders
3. Each order is handled by one salesperson
4. A sales person can handle multiple orders from multiple customers

Q2)Design Schema

No of entities:

1. Customer
2. Order
3. Salesperson

Table Design:

1. Customer(customer_id PK,name,email,phone)
2. Orders(order_id PK,order_date,customer_id FK,salesperson_id FK,amount)
3. Salesperson(salesperson_id PK,name)

Relationship:

1. Customer 1 to many Orders
2. Salespersons 1 to many Orders

Q3)Create Tables

```
mysql> CREATE TABLE Customer (  
->     customer_id INT PRIMARY KEY,  
->     name VARCHAR(100) NOT NULL,  
->     email VARCHAR(100) UNIQUE,  
->     phone VARCHAR(15)  
-> );  
Query OK, 0 rows affected (0.05 sec)  
  
mysql> CREATE TABLE Salesperson (  
->     salesperson_id INT PRIMARY KEY,  
->     name VARCHAR(100) NOT NULL  
-> );  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> CREATE TABLE Orders (  
->     order_id INT PRIMARY KEY,  
->     order_date DATE NOT NULL,  
->     amount DECIMAL(10,2) NOT NULL,  
->  
->     customer_id INT,  
->     salesperson_id INT,  
->  
->     FOREIGN KEY (customer_id)  
->         REFERENCES Customer(customer_id),  
->  
->     FOREIGN KEY (salesperson_id)  
->         REFERENCES Salesperson(salesperson_id)  
-> );  
Query OK, 0 rows affected (0.08 sec)
```

Q4)Insert Sample Data

```
mysql> INSERT INTO Customer (customer_id, name, email, phone) VALUES
-> (1, 'Harsh', 'harsh@example.com', '9876543210'),
-> (2, 'Vaibhav', 'vaibhav@example.com', '9123456780'),
-> (3, 'Ananya', 'an@example.com', '9988776655');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Salesperson (salesperson_id, name) VALUES
-> (101, 'Cors'),
-> (102, 'Mike'),
-> (103, 'Harvey');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Orders (order_id, order_date, customer_id, salesperson_id, amount) VALUES
-> (1001, '2024-01-10', 1, 101, 1500.00),
-> (1002, '2024-01-15', 1, 102, 2200.50),
-> (1003, '2024-01-20', 2, 101, 1800.75),
-> (1004, '2024-01-25', 3, 103, 950.00);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

Q5)Find the sales person have multiple orders.

```
mysql> SELECT
->     s.salesperson_id,
->     s.name,
->     COUNT(o.order_id) AS total_orders
-> FROM Salesperson s
-> JOIN Orders o
->     ON s.salesperson_id = o.salesperson_id
-> GROUP BY s.salesperson_id, s.name
-> HAVING COUNT(o.order_id) > 1;
+-----+-----+-----+
| salesperson_id | name | total_orders |
+-----+-----+-----+
|           101 | Cors |           2 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

Q6)Find the all sales person details along with order details.

```
mysql> SELECT
->     s.salesperson_id,
->     s.name AS salesperson_name,
->     o.order_id,
->     o.order_date,
->     o.amount,
->     o.customer_id
-> FROM Salesperson s
-> LEFT JOIN Orders o
->     ON s.salesperson_id = o.salesperson_id
-> ORDER BY s.salesperson_id;
```

salesperson_id	salesperson_name	order_id	order_date	amount	customer_id
101	Cors	1001	2024-01-10	1500.00	1
101	Cors	1003	2024-01-20	1800.75	2
102	Mike	1002	2024-01-15	2200.50	1
103	Harvey	1004	2024-01-25	950.00	3

4 rows in set (0.00 sec)

Q7)Create index

```
mysql> CREATE INDEX idx_salesperson_id ON Salesperson(salesperson_id);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Q8)How to show index on a table

```
mysql> Show index from Salesperson;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name      | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Salesperson | 0 | PRIMARY      | 1 | salesperson_id | A        | 3 | NULL | NULL |      | BTREE      |         |               | YES     |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Salesperson | 1 | idx_salesperson_id | 1 | salesperson_id | A        | 3 | NULL | NULL |      | BTREE      |         |               | YES     |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

Q9)Find the order number, sale person name, along with the customer to whom that order belongs to

```
mysql> SELECT
->     o.order_id AS order_number,
->     s.name AS salesperson_name,
->     c.name AS customer_name
-> FROM Orders o
-> JOIN Salesperson s
->     ON o.salesperson_id = s.salesperson_id
-> JOIN Customer c
->     ON o.customer_id = c.customer_id;
+-----+-----+-----+
| order_number | salesperson_name | customer_name |
+-----+-----+-----+
|          1001 | Cors             | Harsh         |
|          1003 | Cors             | Vaibhav       |
|          1002 | Mike             | Harsh         |
|          1004 | Harvey           | Ananya        |
+-----+-----+-----+
4 rows in set (0.00 sec)
```